Cross-Variety Initialization and Alignment-Supervised Durations for One-Hour Hakka Speech Synthesis

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***Abstract*—Hakka is a tonal language with several regional varieties, and recorded speech for most of them is scarce. This paper studies neural text-to-speech (TTS) for two Taiwanese Hakka varieties, Sixian and Hailu, when only one hour of target-variety speech is available. With so little data, a TTS model that learns its own alignment produces imprecise segment timing and weak tones. The proposed XAD-TTS is a training recipe for Matcha-TTS that initializes the model from a related source variety and supervises an enlarged duration predictor with forced-alignment durations. Across three training seeds, XAD-TTS raises the share of checked-syllable vowels whose duration, measured on the synthesized waveform, lies within 30 ms of natural speech from 71.0% to 81.3% (Sixian) and from 69.4% to 82.2% (Hailu), on par with a baseline trained on about 48 times more data. It also lowers mel-cepstral distortion by 0.33 and 0.24 dB and raises objective tone recognition by 8.8 and 5.3 points. An independent check with a multilingual speech recognizer confirms that checked syllables and tone marks are better preserved.**

***Index Terms*—duration modeling, forced alignment, Hakka, low-resource languages, speech synthesis, transfer learning**

I. Introduction

Hakka is a Sinitic tonal language. In Taiwan it is spoken in several regional varieties, two of the major ones being Sixian and Hailu. The two share most consonants and vowels but differ in their tone systems. Speech technology for Hakka has begun to appear [1], yet the recorded speech available for each variety is limited, and a voice for a new variety or speaker often has to be built from very little data.

Neural TTS models such as Matcha-TTS [2] learn the alignment between input symbols and acoustic frames during training, typically with monotonic alignment search (MAS) [3], and train a duration predictor on the resulting durations. With one hour of data, both the learned alignment and the predictor are noisy. In Hakka this affects more than naturalness: checked syllables, the syllables of the entering tones that end in an unreleased stop (written *-b*, *-d*, *-g*), are marked by a short vowel, so segment timing carries a phonological contrast.

Two ideas have been used separately for low-resource TTS: initializing from a model of a related, better-resourced language [4], [5], and replacing learned alignments with durations from an external aligner, as FastSpeech 2 [6] does with the Montreal Forced Aligner (MFA) [7]. This paper combines both for Hakka and examines how far one hour of target-variety data can go. The contributions are as follows:

- **XAD-TTS**, a training recipe that initializes the target model from a related source variety and supervises an enlarged duration predictor with forced-alignment durations. The aligner is needed only for training, and no component is added at inference.
- Controlled experiments on two varieties, three data sizes (1, 10, and about 48 h), and three training seeds, with segment timing measured on the synthesized waveforms.
- An evaluation beyond spectral distance, comprising objective tone recognition and an independent check with a multilingual speech recognizer.

II. Method

A. Baseline model

The baseline, denoted Base, is the original Matcha-TTS [2] trained from scratch. It encodes the input symbols into per-symbol features and log-durations, upsamples the features by the durations, and generates a mel spectrogram with a conditional flow-matching decoder [8]; a HiFi-GAN vocoder [9] produces the waveform. During training, MAS chooses the durations that best explain the mel frames under the encoder output, and the duration predictor is fitted to them. The training loss is the sum of the duration, prior, and flow-matching losses. At inference, predicted durations are rounded up to whole frames.

B. Cross-variety initialization

The source model is trained from scratch on the full training pool of the other variety (48.0 h of Sixian or 47.2 h of Hailu); Sixian and Hailu are closely related varieties of Hakka. Both varieties use one symbol table, so every target parameter with a matching name and shape is copied from the source model. Speaker embeddings are re-initialized, no parameter is frozen, and the optimizer starts afresh at step 0. Because the copied

mel normalization statistics belong to the source variety, synthesis always uses the statistics of the target training data.

C. Alignment-supervised durations

An MFA acoustic model and a pronunciation lexicon were trained on the corpus, and every training utterance was aligned. Phone intervals are converted to frame counts (hop 256 at 22.05 kHz, i.e., 11.6 ms per frame). Matcha-TTS inserts a blank token between symbols; aligned pauses are assigned to the corresponding blank, and every blank receives at least one frame, taken from the preceding phone. These durations replace MAS: they are the targets of the duration loss and define the upsampling used by the prior and decoder losses.

D. Larger duration predictor and rounding

To give the duration predictor more capacity, it is enlarged from kernel size 3 and 256 channels to kernel size 5 and 512 channels. With cross-variety initialization, the enlarged predictor and the speaker embeddings are the only re-initialized parts. Rounding every predicted duration up adds about half a frame per symbol on average, which can lengthen segments when the predictor itself is unbiased. Each trained model therefore uses either rounding up or cumulative rounding (rounding the running sum, which preserves the total length), whichever gives vowel durations closer to the training targets on the validation set. The same selection procedure is applied to all systems, including the baseline.

XAD-TTS combines Sections II-B to II-D. Fig. 1 shows how it is trained, used, and evaluated. The target model starts from the source-variety model through cross-variety initialization, and the MFA durations of the one-hour target training speech supervise its enlarged duration predictor. After training, the model synthesizes speech from test text, and the synthesized speech is evaluated against natural test speech.

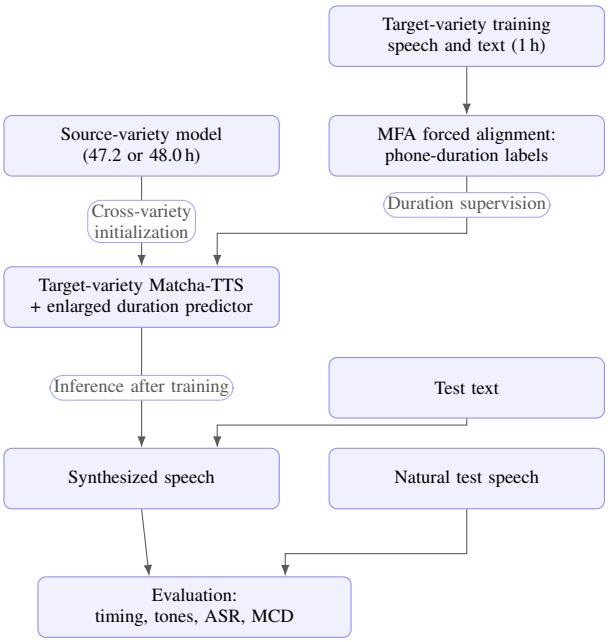


Fig. 1. Training, synthesis, and evaluation of XAD-TTS.

III. Experimental Setup

A. Data

The experiments use the TTS part of the Hakka Across Taiwan (HAT) corpus [10]: read speech in Sixian and Hailu, each by one professional female and one professional male speaker. Validation and test sets are defined by recording prompt, so their prompts never occur in training. Training subsets of 1 h and 10 h are balanced across the two speakers and nested (the 10 h set contains the 1 h set). Pinyin is split deterministically into initial, vowel nucleus, and coda, and the tone is attached to the nucleus symbol ("joint" symbols). Section IV-A also uses "separate" symbols, in which tones are separate inputs, so 84% instead of 55% of the symbols are shared by the two varieties. Over 99% of the utterances were aligned successfully.

B. Systems and training

Three systems are compared, all with joint symbols: Base, the original Matcha-TTS baseline trained from scratch with MAS durations; +AD, trained from scratch with alignment-supervised durations; and XAD, which adds cross-variety initialization and the enlarged predictor to +AD. Each is trained with Adam (learning rate 10^{-4}, batch 16) for a fixed 40,000 steps with three training seeds (1234, 123, 2026) at 1 h and 10 h, and the 40,000-step checkpoint is evaluated. As a high-resource reference, Base is also trained on the full pool (48.0 h of Sixian, 47.2 h of Hailu) with early stopping and a single seed. One 40,000-step run takes about 2 h on a single RTX 5070 Ti GPU (about 5.3 steps/s, 3.9 GB memory), and Base has 20.9 M parameters. Synthesis uses 10 ODE steps, temperature 0.667, and the universal HiFi-GAN vocoder.

C. Evaluation

For each variety, 100 test sentences (50 per speaker) are synthesized. 95% confidence intervals (CIs) are obtained by resampling sentences 5,000 times; with several seeds, every seed uses the same resampled sentences.

Spectral distortion. Mel-cepstral distortion (MCD) [11] is computed on 24 WORLD [12] mel-cepstral coefficients (without c_0) after dynamic time warping and averaged over five generation seeds per sentence.

Segment timing. Each synthesized waveform is force-aligned with the same MFA model, and the duration of every vowel nucleus is compared with the same nucleus in the natural recording. Timing accuracy is the share of nuclei within ± 30 ms (2.6 frames) of the natural duration, reported separately for checked and other syllables, together with the Pearson correlation ρ of checked-syllable nucleus durations. Tolerances of ± 20 and ± 50 ms give the same ranking of Base, +AD, and XAD.

Tone recognition. A gradient-boosting classifier is trained on natural training speech with acoustic features only: a five-point F0 contour of the nucleus in semitones, its mean and slope, log-duration, voicing ratio, and the F0 of neighboring syllables. Its accuracy on the synthesized syllables is reported,

and its accuracy on natural test recordings serves as a reference.

Speech recognition check. Omnilingual ASR [13] (1B LLM decoder, version 2, language code `hak_Latn`) transcribes Hakka in a romanization with tone diacritics and stop codas *-p*, *-t*, *-k*. Each natural recording and its synthetic versions are transcribed, and the transcripts are aligned word by word. The measures are the character error rate (CER) of the synthetic transcript against the natural one, the retention of checked syllables (natural-transcript words ending in *-p*/*-t*/*-k* whose aligned synthetic word still does), the letter-similarity gap between checked and other words, and the tone-mark agreement of aligned words. Twenty of the 200 test sentences are excluded because the recognizer's output for the natural recording was mostly non-Latin text, and the results are pooled over the two varieties. Copy-synthesis, in which natural mel spectrograms are passed through the same vocoder, serves as an upper bound.

IV. Results

A. Cross-variety initialization

In a preliminary sweep over five data sizes from 1 to 48 h, initialization lowered the validation loss in 16 of 20 settings; the gain was largest at 1 h (0.08–0.13) and vanished at 48 h, so the analysis focuses on 1 h. Table I compares MCD at 1 h between training from scratch and cross-variety initialization under a controlled protocol: 40,000 steps, best validation checkpoint, and one training seed, with the 95% CI of the paired difference in brackets.

TABLE I
MCD at 1 h with and without cross-variety initialization.

Target	Symbols	Scratch	Init.	Δ [95% CI]
Sixian	joint	6.36	5.99	−0.36 [−0.39, −0.34]
Sixian	separate	6.17	6.08	−0.09 [−0.12, −0.06]
Hailu	joint	6.46	6.59	+0.13 [+0.09, +0.18]
Hailu	separate	6.48	6.22	−0.26 [−0.30, −0.22]

Initialization lowers MCD in three of four settings, by up to 0.36 dB, but raises it for Hailu with joint symbols (+0.13 dB), mostly for the male speaker (+0.35 dB vs. −0.08 dB). Initialization alone is therefore helpful but not uniformly so, and XAD-TTS pairs it with alignment supervision.

B. Main results at one hour

Table II summarizes the results at 1 h. Checked and Other give the timing accuracy of checked and other syllables, ρ the duration correlation of checked syllables, ΔMCD the paired difference from Base (1 h), and Tone the tone recognition accuracy, all averaged over three training seeds. Base (48.0 h) and Base (47.2 h) are the high-resource references, and Natural recordings gives the accuracy of the tone classifier on real speech.

TABLE II
Results at 1 h for each system and variety.

System	Checked (%) ↑	Other (%) ↑	ρ ↑	ΔMCD (dB) ↓	Tone (%) ↑
Sixian					
Base (1 h)	71.0	63.7	0.535	—	64.3
+AD (1 h)	77.0	70.4	0.651	−0.12	69.0
XAD (1 h)	**81.3**	**74.0**	**0.738**	**−0.33**	**73.0**
Base (48.0 h)	81.1	73.9	0.767	—	—
Natural recordings	—	—	—	—	71.5
Hailu					
Base (1 h)	69.4	54.9	0.576	—	62.1
+AD (1 h)	77.7	58.9	0.676	−0.10	64.2
XAD (1 h)	**82.2**	**65.9**	**0.743**	**−0.24**	**67.3**
Base (47.2 h)	83.0	68.6	0.797	—	—
Natural recordings	—	—	—	—	67.1

Both components help, and their gains add up. Alignment supervision alone (+AD) raises the share of checked-syllable vowels within ±30 ms by 6.0 (Sixian) and 8.3 (Hailu) points. XAD adds another 4.3 and 4.5 points, reaching 81.3% [77.6, 85.2] and 82.2% [78.0, 86.4] against 71.0% [66.9, 75.2] and 69.4% [65.3, 73.8] for Base, which matches the high-resource references (81.1% and 83.0%). Other syllables improve as well (63.7% to 74.0% and 54.9% to 65.9%), and the correlation of checked-syllable durations rises from 0.54 to 0.74 (Sixian) and from 0.58 to 0.74 (Hailu), still below the references (0.77 and 0.80). XAD lowers MCD by 0.33 [0.31, 0.35] and 0.24 [0.20, 0.28] dB, two to three times the reduction from +AD (0.12 and 0.10 dB), and all three seeds move in the same direction. Tone recognition rises by 8.8 [7.7, 9.9] and 5.3 [4.4, 6.2] points, and XAD exceeds +AD by 4.1 and 3.2 points. XAD's syllables are classified about as accurately as natural recordings (measured on all test sentences); this more likely reflects less variable, more canonical tones than tones that are better than natural ones.

C. Effect of data size

Fig. 2 compares checked-syllable timing accuracy at 1 h and 10 h, with three seeds each, against the single high-resource Base model; the 95% CIs of the plotted values are about ±4 points.

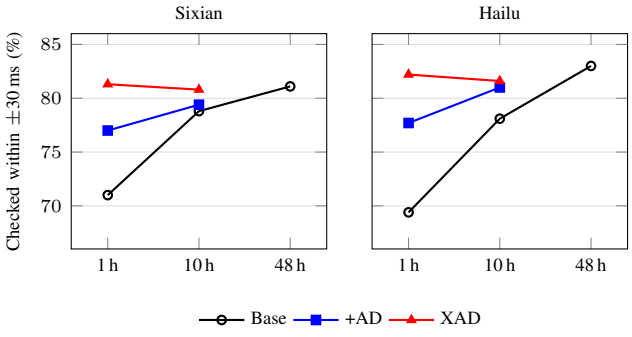


Fig. 2. Checked-syllable timing accuracy versus amount of training data.

The gains shrink with more data. At 10 h, XAD improves the checked-syllable share by 2.0 (Sixian) and 3.5 (Hailu)

points and MCD by 0.24 and 0.16 dB, while +AD alone no longer lowers MCD (+0.04 and +0.03 dB). The recipe is therefore most useful in the regime it targets, when only about an hour of speech is available; there, one hour with XAD matches the checked-syllable timing of a baseline trained on 48 times more data.

D. Speech recognition check

Table III lists the speech recognition measures pooled over both varieties (180 sentences), including the copy-synthesis upper bound; Gap is the letter similarity of checked words minus that of other words.

TABLE III
SPEECH RECOGNITION CHECK OF THE SYNTHESIZED SPEECH.

System	CER (%) ↓	Checked kept (%) ↑	Gap ↑	Tone marks (%) ↑
Base (1 h)	22.2	77.8	−4.1	73.9
+AD (1 h)	21.8	81.5	−1.5	76.7
XAD (1 h)	21.7	**82.6**	**−0.5**	77.1
Base (47.2/48.0 h)	19.6	80.1	−3.7	77.2
Copy-synthesis	9.8	89.4	+0.2	88.9

The recognizer never saw the aligner or the tone classifier, so it gives an independent view. Overall CER does not separate the 1-h systems (−0.5 [−2.7, +1.9] points for XAD), while the high-resource baseline is better (−2.6 [−4.5, −0.5]). The differences appear where XAD targets them. Checked syllables keep their stop coda more often (+4.8 [1.6, 8.0] points), the handicap of checked words relative to other words shrinks from −4.1 to −0.5 points (+3.6 [1.2, 6.0]), and tone marks agree more often with those of natural speech (+3.2 [1.9, 4.4]). The high-resource baseline improves tone marks equally but leaves the checked-word handicap almost unchanged (−3.7), which suggests that the checked-syllable gain comes from duration supervision rather than from more data. With +AD alone the same pattern holds with smaller gains (+3.7, +2.6, and +2.8 points). Copy-synthesis reaches 89.4% retention, a gap of +0.2, and 88.9% tone agreement. XAD thus removes the checked-word handicap and closes about 40% of the retention gap between Base and this bound, while the large CER difference (9.8% vs. 21.7%) shows that most of what remains is general acoustic quality rather than timing.

E. Negative results

Inference-time duration scaling, dropout in or pre-training of the duration predictor, replaying source-variety data during target training, and averaging the durations of several seeds gave no further gain over XAD.

V. Discussion and Limitations

At one hour, alignment supervision accounts for most of the timing gain (6.0 of 10.3 points for Sixian, 8.3 of 12.8 for Hailu), while cross-variety initialization with the larger predictor contributes most of the spectral gain and about half of the tonal gain. The remaining vowel-duration errors concentrate before pauses, where the predictor lengthens vowels more than the speakers do. The study has limitations. A listening test has not yet been completed, so all results are objective; a paired-comparison listening test is under way. All systems share one architecture and vocoder. The recipe needs a pronunciation lexicon and an in-domain aligner. The high-resource reference is a single model. The speech recognizer and the tone classifier are imperfect judges; the classifier reaches only 71.5% and 67.1% on natural speech.

VI. Conclusion

XAD-TTS combines cross-variety initialization with alignment-supervised durations and an enlarged duration predictor. With one hour of Hakka speech, it brings checked-syllable timing to the level of a baseline trained on 48 hours, reduces spectral distortion, and improves objective tone recognition in both Sixian and Hailu, and an independent multilingual recognizer confirms the gains on checked syllables and tones. The recipe adds no inference-time component and trains in about two hours on a single consumer GPU, which makes it a practical starting point for voices in other low-resource varieties.

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